



If we compare the Indian design ecosystem with the global standard, what do we need to become a design powerhouse?

Certainly, if we observe companies in the technology sector, much of the design work occurs in India, indicating a strong talent base in design. However, Indian-owned businesses in this sector are relatively few. Currently, most companies operating in India are either US or UK-owned, or they are European companies with a presence in India. Indian designers often work for these companies, developing designs that are sold internationally. To change this, we need to focus on building Indian-owned companies capable of selling their IPs and products globally, rather than limiting their market to India. The challenge lies in enabling that and making those resources more affordable to encourage more people to enter the industry early on. Otherwise, there is a risk of a heavy reliance on Python programming, with fewer advancements in hardware design originating from India.

What role does Digital University Kerala (DUK) play here?

With the implementation of a model, we have completely removed the need for traditional classroom-based teaching and transitioned to experiential learning. This means there are no fixed learning hours; it is more about continuous learning, accessible 24/7. Our curriculum is designed around lab-based and project-based learning. This shift is driven by industry demands for targeted skills rather than generic ones. Our curriculum, designed around lab-based and project-based learning, is driven by the demand for industry-specific skills. This model encourages problem-solving and creativity by removing reliance on textbooks, focusing instead on user manuals, design guides, and

reference standards.

Additionally, we have established several independent centres of excellence, often registered as companies, to support this ecosystem. These centres include incubators and research facilities focused on commercial product development. Many students participate in these centres while pursuing their master's or PhDs, leading to the creation of start-up companies, particularly in the electronic design space. In the last two years alone, we have seen the emergence of around four start-up companies, largely driven by students.

In partnership with IEEE, we have launched a programme called Maker CHIPS to democratise IC design using open source tools. Students and researchers get a chance to learn the art of IC design and manufacture the chips at no cost to the end user. The details of this programme can be found at www.makerchips.org.

What is the agenda behind encouraging and promoting these start-ups—is it to build a major innovative player in the ecosystem or create a profitable, revenue-generating business, which can be acquired by an investor or a larger company?

The first agenda is to cultivate an entrepreneurial culture among universities. In our educational system, there is a prevalent focus on grades and securing placements. To shift this mindset, we emphasise problem-solving in real-world contexts, understanding customer needs, market conditions, and commercial research interests. This approach changes students' attitudes towards research and the realities of the world. Entrepreneurial thinking is mandatory for PhD and master's students, driving them to engage with actual problems and customers as part of their graduation

requirement. By imposing these constraints, we see a shift in their worldview and interactions with others. Additionally, we offer courses that encourage creativity, problem-solving, and practical skills, such as drama, storytelling, finance for non-financial managers, and IP management. All of this is part of an electronics VLSI programme. Community engagement and empowerment courses expose students to real societal issues, encouraging them to design solutions that address these problems, fostering entrepreneurial thinking and opening up new possibilities for innovation.

So, what does that mean—are you prioritising innovation or business opportunity?

The focus is on innovation and entrepreneurship, but success in business goes beyond just creating a start-up. It is about developing meaningful and useful products, effectively marketing and selling them, and ultimately building profitable businesses that can scale. While scaling may require manufacturing support and significant investment, the Indian venture ecosystem is still nascent, especially in hardware. Our goal is to enable start-ups to find their market and attract investment, regardless of where it comes from. By instilling confidence in entrepreneurs to seek investment globally, we aim to help them take their ventures to the next level. While not every start-up may reach that level of success, even having one or two succeed would be a significant achievement. Our efforts are focused on nurturing this mindset and supporting start-ups as they strive for growth and success.

What happens when a start-up comes up with an idea that sells well but has a lot of established players in the market doing the